

1. Operating point analysis. This analysis simply calculates the DC operating point. No parameters are required.

2. DC transfer curve analysis. In this analysis, one or two sources—a voltage or a current source—are swept over a user-defined interval. DC operating point of the circuit is calculated for each value of the source(s). DC transfer curve analysis is useful in finding the logic swing of logic gates or I-V characteristics of a transistor for example.

3. AC small signal analysis. This analysis calculates the frequency response of a circuit by linearising circuit equations around the operating point.

4. Transient analysis. The time domain response of a circuit is calculated from $t=0$ to a user-defined upper time limit. This analysis has two optional parameters. The first one specifies that the generation of output starts at a value different from zero. The second optional parameter sets an upper limit on the time steps used by AIM-Spice.

5. Pole-zero analysis. AIM-Spice is able to locate poles and zeroes in an AC small signal transfer function. First, DC operating point is calculated, and then the circuit is linearised around the operating point. The resulting circuit is used to locate poles and zeroes.

6. Transfer function analysis. This analysis computes the DC small signal value of the transfer function, input resistance and output resistance.

7. Noise analysis. Noise analysis portion of AIM-Spice computes device-

First Example

```
* divider01.ckt - Your Name - Date
* A simple divider example
```

```
R1 1 2 1K
R2 2 0 1K ! Node zero is ground
V1 1 0 DC 10V
```

```
* end of divider01.ckt
```

The text description on the left describes the same circuit as the schematic on the right

Components are listed and are interconnected in networks by nodes

Node 0 refers to ground

There must be a DC path from every node to ground

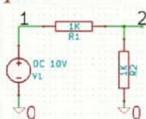


Fig. 2: Example 1

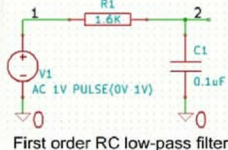
Second Example

```
* lowpass01.cir - Your name - date
```

```
* A first order low-pass RC filter
```

```
R1 1 2 1.6K
C1 2 0 0.1uF ! 'IC = val' for initial cond.
V1 1 0 AC 1V PULSE(0V 1V)
```

```
* end of lowpass01.cir
```



First order RC low-pass filter

Fig. 3: Example 2

generated noise for the given circuit. When provided with an input source and an output port, it calculates noise contributions of each device and each noise generator within the device to the output port voltage. It also calculates input noise to the circuit, equivalent to output noise referred to the specific input source. This is done for every frequency point in a specified range.

8. Distortion analysis. Small signal distortion analysis part of AIM-Spice calculates steady-state harmonic and inter-modulation products for small signal input magnitudes. If signals of

a single frequency are given as input to the circuit, values of the second and third harmonics are computed. If there exist input signals of two frequencies, the analysis finds out values of circuit variables at sum and difference of the input.

All analyses are made available as commands from the Analysis menu. All analysis commands, except for the DC operating point analysis, require additional control parameters to be specified. However, before choosing one of the commands in the Analysis menu, one should decide which circuit to analyse and make the corresponding circuit window the active window.

New semiconductor device models

For the following semiconductor devices, new advanced models are implemented in AIM-Spice:

1. Heterostructure diode
2. Heterojunction bipolar transistor
3. MOSFETs. The following MOSFET models are included:

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- Level 1, 2, 3 and 6 from Berkeley
- BSIM1
- BSIM2
- BSIM3v2
- BSIM3v3.1
- BSIM3v3.2
- BSIM3 SOI
- BSIM4 versions 1, 2, 2.1, 3 up to 6
- 4. GaAs MESFETs and HFTs
- 5. Amorphous silicon thin-film transistors (a-Si TFTs)
- 6. Polysilicon thin-film transistors (poly-Si TFTs)

All the FET models are based on